

6.7980: Topics in Multiagent Learning

Lecture 1 – Prof. Daskalakis
Fall 2026



The Concept of Game

Games are thought experiments to help us learn how to *predict rational behavior* in *situations of conflict*.

Situation of conflict: Everybody's actions affect others.

Rational Behavior: The players want to maximize their own expected utility. No altruism, envy, masochism, or externalities (if my neighbor gets the money, he will buy louder stereo, so I will hurt a little myself...).

Predict: We want to know what happens in a game. Such predictions are called *solution concepts* (e.g., Nash equilibrium).

Situations Modeled as Games

- **Recreational games**

- Rock paper scissors
- Diplomacy
- Poker
- Go
- ...

- **Non-recreational settings**

- Auctions
- Markets
- Logistics
- Budget allocation (e.g., political campaigns)
- Generative adversarial networks
- Multi-robot interactions
- Fraud detection systems
- Cyber-defense
- Agentic AI
- ...

eg1: Prisoner's Dilemma

Setting: Two members of a criminal gang are arrested and imprisoned. Each prisoner is in solitary confinement with no means of speaking to or exchanging messages with the other.

The police admit they don't have enough evidence to convict the pair on the principal charge. They plan to sentence both to a year in prison on a lesser charge.

Simultaneously, the police offer each prisoner a bargain:

- If A and B both provide evidence against each other, each of them serves 2 years in prison
- If A provides evidence against B but B does not, A will be set free whereas B will serve 3 years in prison (and vice versa)
- If neither A nor B provide evidence against the other, both will serve 1 year

	don't betray	betray
don't betray	-1 -1	-3 0
betray	0 -3	-2 -2

(negative numbers as these are years in prison so negative utilities)

eg1: Prisoner's Dilemma

	don't betray	betray
don't betray	-1 -1	-3 0
betray	0 -3	-2 -2

What do we expect to happen?

- From purple player's point of view, **betray dominates don't betray**, i.e. no matter what green plays purple is better off by playing **betray**.
- Likewise, From blue's point of view, **betray dominates don't betray**.
- So the rational strategy for both is to play betray.
- It is a **dominant strategy equilibrium**.
- It is worse for both compared to (**don't betray, don't betray**)

eg2: Rock-Paper-Scissors

	rock	paper	scissors
rock	0 0	-1 1	1 -1
paper	1 -1	0 0	-1 1
scissors	-1 1	1 -1	0 0

- No dominant strategy equilibrium.
- **Next compromise:** Prepare for the worst?
 - Captured by solving: $\arg\max_x \min_y (-x_r y_p + x_r y_s + x_p y_r - x_p y_s - x_s y_r + x_s y_p)$
 - optimal solution: $x^* = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$
 - By symmetry, if **column** player wants to prepare for the worst: $y^* = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$
 - Cool coincidence: x^* is optimal against y^* and vice versa
 - coincidence holds in any zero-sum game [von Neumann's minimax theorem]

eg3: Battle of the Sexes

	Theater!	Football fine
Theater fine	1 5	0 0
Football!	0 0	5 1

- No dominant strategy equilibrium.
- Let us try max min?
 - pessimistic strategy for row player: $\operatorname{argmax}_x \min_y (x_t y_t + 5x_f y_f) \rightarrow x^* = \left(\frac{5}{6}, \frac{1}{6}\right)$
 - pessimistic strategy for column player: $\operatorname{argmax}_y \min_x (5x_t y_t + x_f y_f) \rightarrow y^* = \left(\frac{1}{6}, \frac{5}{6}\right)$
 - Issue: x^* is NOT optimal against y^* and y^* is NOT optimal against x^*

Question: Do there exist strategies x and y such that x is optimal against y and vice versa?

eg3: Battle of the Sexes

		5/6	1/6
		Theater!	Football fine
1/6	Theater fine	1 5	0 0
5/6	Football!	0 0	5 1

Question: Do there exist strategies x and y such that x is optimal against y and vice versa?

The answer is yes, and it is captured by the concept of **Nash Equilibrium**: A pair of strategies (deterministic or randomized) such that the strategy of the row player is at least as good as any other strategy given the strategy of the column player, and vice versa.

- (Theater fine, Theater!) “Green-archic society”
- (Football! , Football fine) “Purple-archic society”

Teaser: There always exists an odd number of Nash Equilibria in two-player games

Discussion

All the games we have considered are **one-shot games of complete information**.

- “complete information:” Players also know everything about the payoffs of each other.
- “one shot:” Players meet for one interaction and that is it; there are no stages/sequential decisions

One-shot games intend to model repeated occurrences of the same conflict, provided that there are no strategic correlations between different occurrences of the game. If such correlations exist, we exit the realm of one-shot games, entering the realm of **repeated games**.

How can repeated occurrences happen without inter-occurrence correlations?

Imagine a population of **purple players** (these are the ones preferring football) and **green players** (these are those preferring theater). Members of the **purple** population encounter members of the **green** population at random and need to decide whether to watch football or theater.

*What are the **Nash equilibria** supposed to capture?*

Nash equilibria predict what types of behaviors and (in the case of randomized strategies) at what proportions will arise in the two populations in the steady state.

More Abstract Game Formulation

- **Def:** A *finite game (in normal-form)* is described by:
 - a set of players $[n] = \{1, \dots, n\}$
 - a set of *pure strategies/actions* for each player $i \in [n] : A_i$
 - a *utility/payoff function* for each player $i \in [n] : u_i : A_1 \times \dots \times A_n \rightarrow \mathbb{R}$
- **Def:** A *randomized/mixed strategy* for player i is any $x_i \in \Delta(A_i)$
 - assigns probability $x_i(a_i)$ to each $a_i \in A_i$
 - i.e. $\Delta(A_i)$ is the simplex whose vertices are identified with the elements of A_i
- **Def:** a player's *expected utility* is
 - $u_i(x_1, \dots, x_n) = \mathbb{E}_{a_1 \sim x_1, \dots, a_n \sim x_n} [u_i(a_1, \dots, a_n)] \equiv \sum_{a \in \times_j A_j} u_i(a) x_1(a_1) \cdot \dots \cdot x_n(a_n)$
- A piece of very useful notation: if $v = (v_1, \dots, v_n)$ is a collection of n objects then v_{-i} denotes the same collection excluding object v_i
- **Def:** a collection of mixed strategies $x_1, \dots, x_n$ is a *Nash equilibrium* iff
 - $\forall i, x'_i : u_i(x_i, x_{-i}) \geq u_i(x'_i, x_{-i})$
- **Def:** a collection $x_1, \dots, x_n$ is a *dominant strategy equilibrium* iff
 - $\forall i, x'_i, x'_{-i} : u_i(x_i, x'_{-i}) \geq u_i(x'_i, x'_{-i})$

More Abstract Game Formulation

- **Def:** a collection $x_1, \dots, x_n$ is a *Nash equilibrium* iff
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- **Def:** a collection $x_1, \dots, x_n$ is a *dominant strategy equilibrium* iff
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	Theater!	Football fine
Theater fine	1, 5	0, 0
Football!	0, 0	5, 1

$$x_{blue} = \left(\frac{1}{6}, \frac{5}{6}\right) \quad x_{orange} = \left(\frac{5}{6}, \frac{1}{6}\right)$$

$$\left. \begin{aligned} u_{blue}(x_{blue}, x_{orange}) &= \frac{1}{6} \cdot \frac{5}{6} \cdot 1 + \frac{5}{6} \cdot \frac{1}{6} \cdot 5 = \frac{5}{6} \\ u_{blue}('theater\ fine', x_{orange}) &= \frac{5}{6} \cdot 1 + \frac{1}{6} \cdot 0 = \frac{5}{6} \\ u_{blue}('football!', x_{orange}) &= \frac{5}{6} \cdot 0 + \frac{1}{6} \cdot 5 = \frac{5}{6} \end{aligned} \right\} u_{blue}(x_{blue}', x_{orange}) = \frac{5}{6}, \forall x_{blue}'$$

Nash's Theorem



[Nash 1950]: Every finite game (i.e. with a finite number of players and a finite number of pure strategies per player) has a Nash equilibrium.

- We'll prove it!
- We'll make use of **Brouwer's fixed point theorem**, following a proof that Nash produced in 1951; his original proof used Kakutani's fixed point theorem.

Menu

- **Game-theoretic formalism**
- Nash's theorem

Menu

- **Game-theoretic formalism**
- **Nash's theorem**



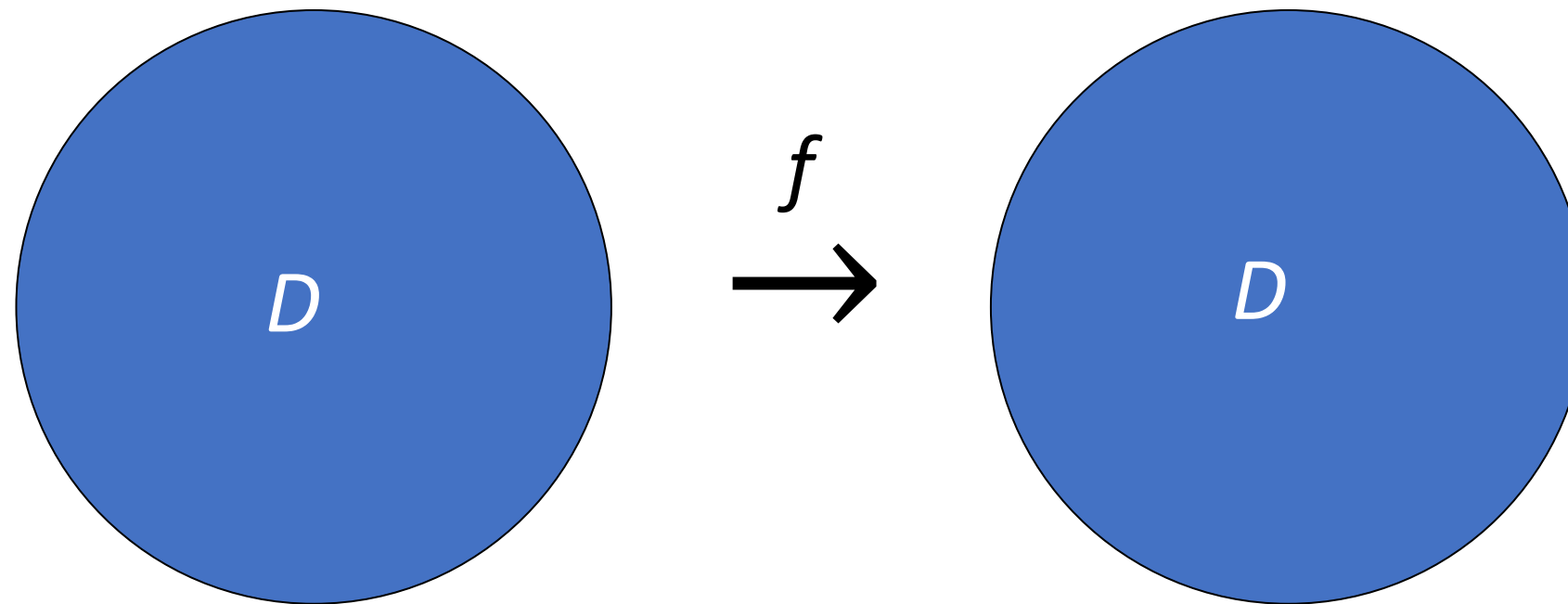
Brouwer's Fixed Point Theorem

[Brouwer 1910]: Let $f : D \rightarrow D$ be a continuous function from a convex and compact subset D of the Euclidean space to itself. Then there exists an $x \in D$ s.t. $x = f(x)$.



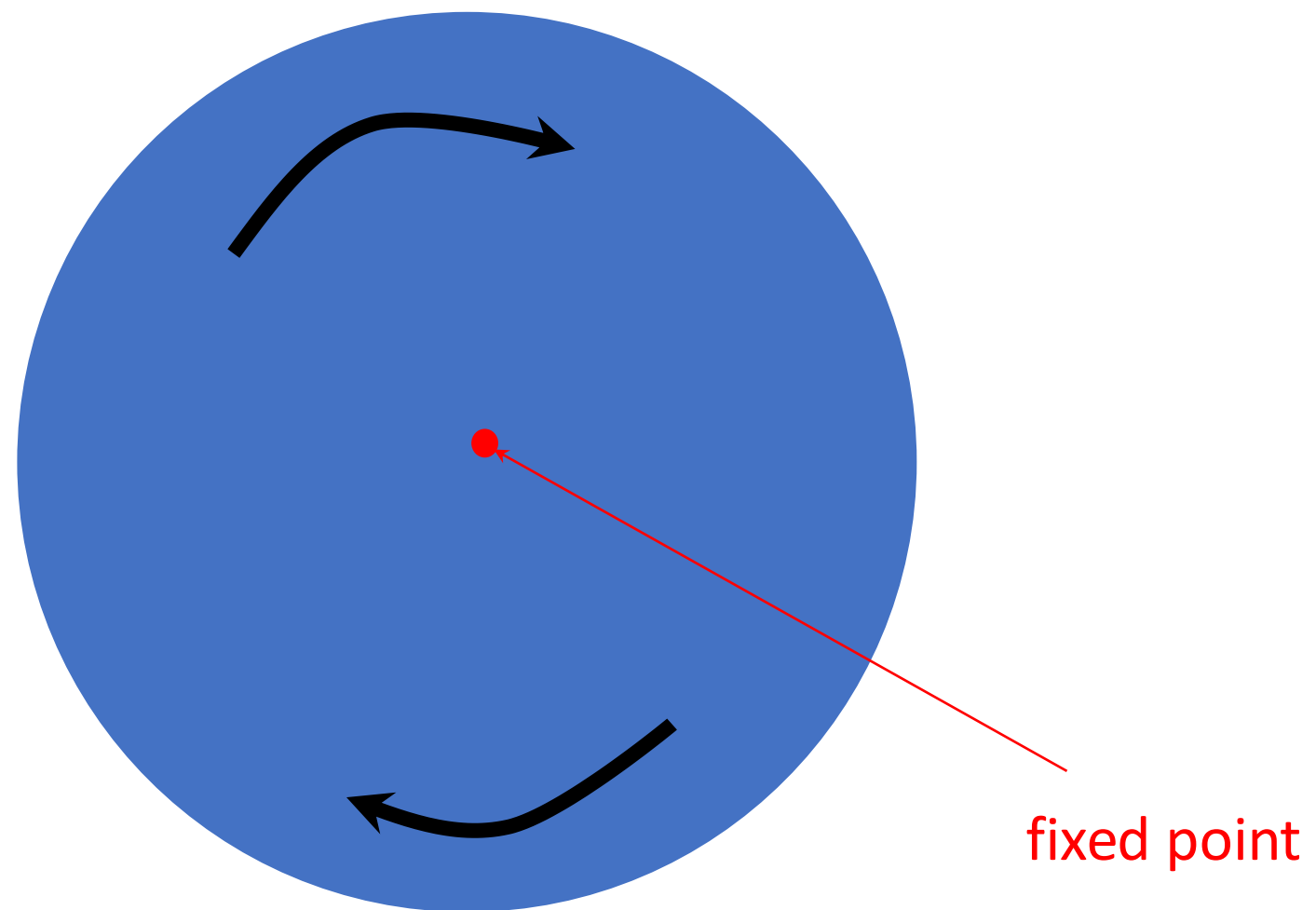
closed and bounded

Below we show a few examples, when D is the 2-dimensional disk.

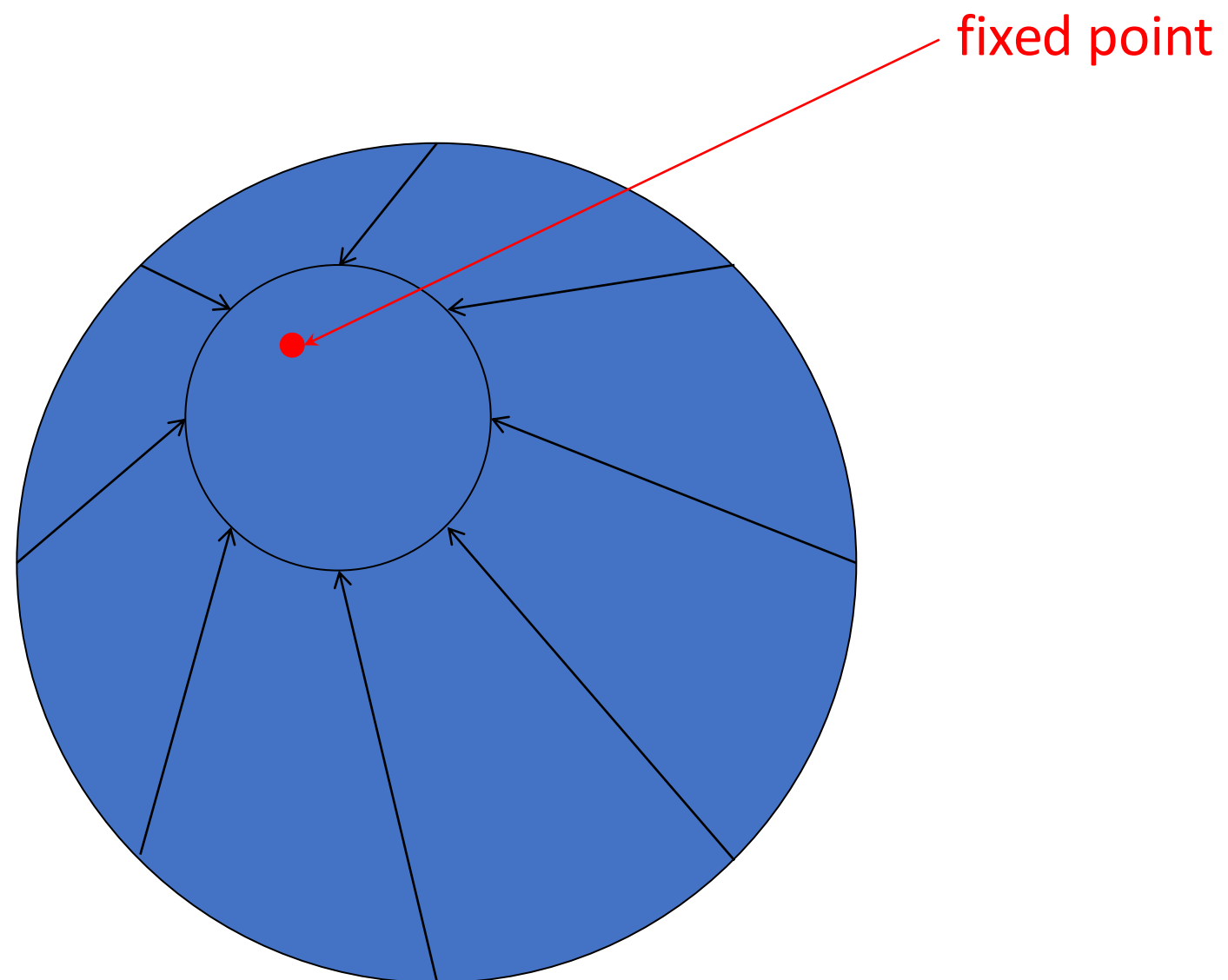


N.B. All conditions in the statement of the theorem are necessary.

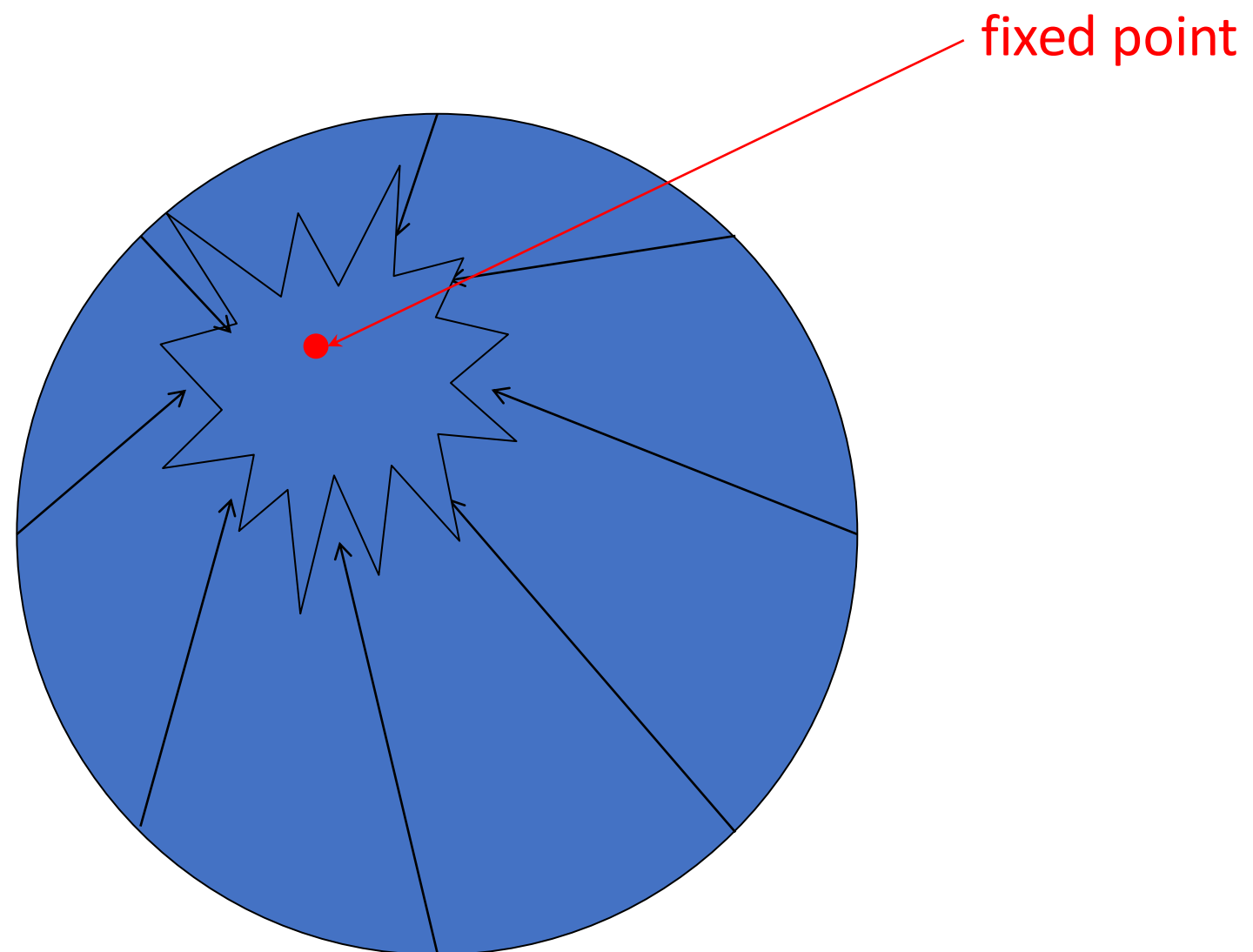
Brouwer's Fixed Point Theorem

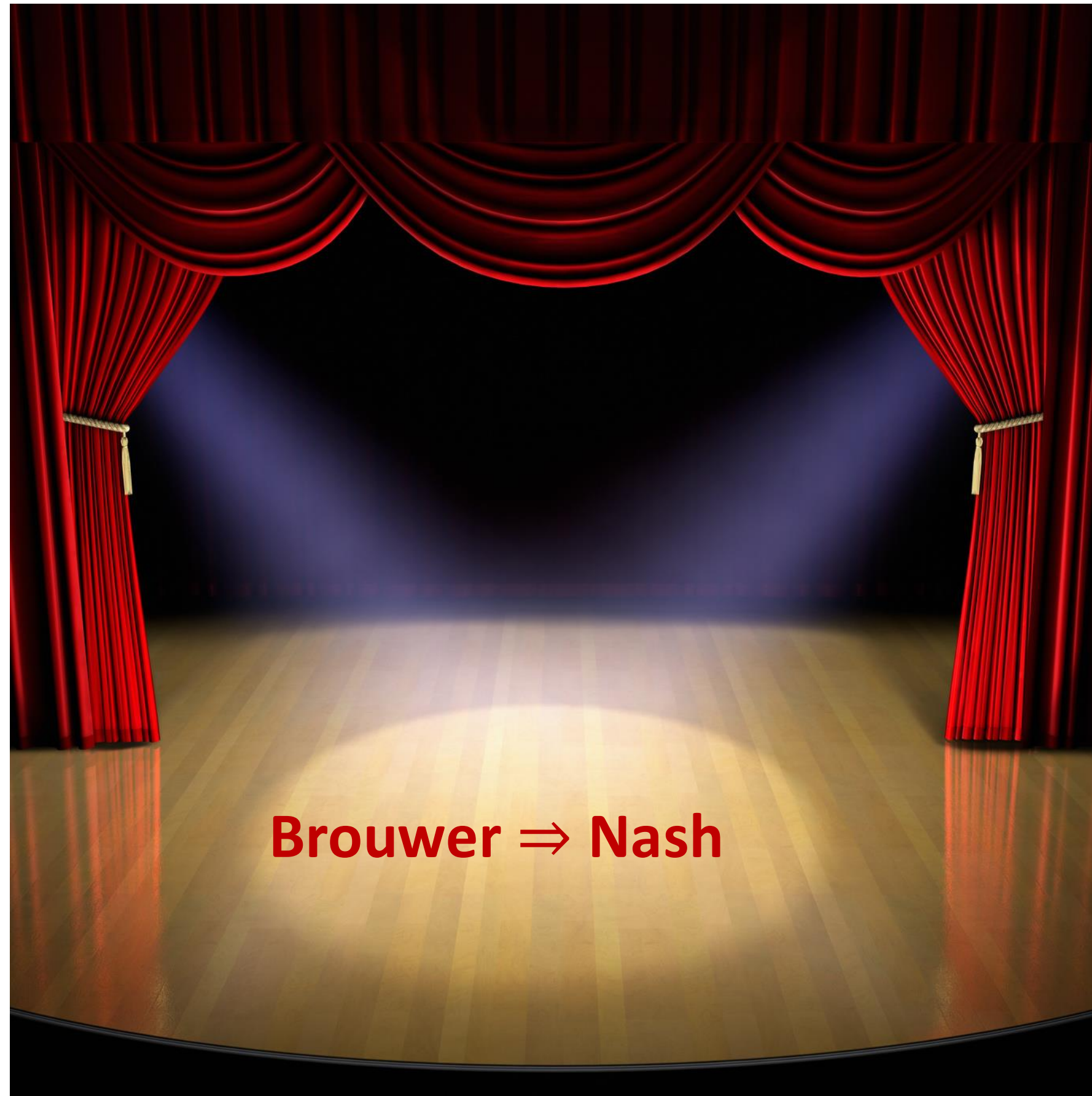


Brouwer's Fixed Point Theorem



Brouwer's Fixed Point Theorem





Brouwer $\Rightarrow$ Nash

Visualizing Nash's Proof

Kick Dive	Left	Right
Left	1 , -1	-1 , 1
Right	-1 , 1	1 , -1

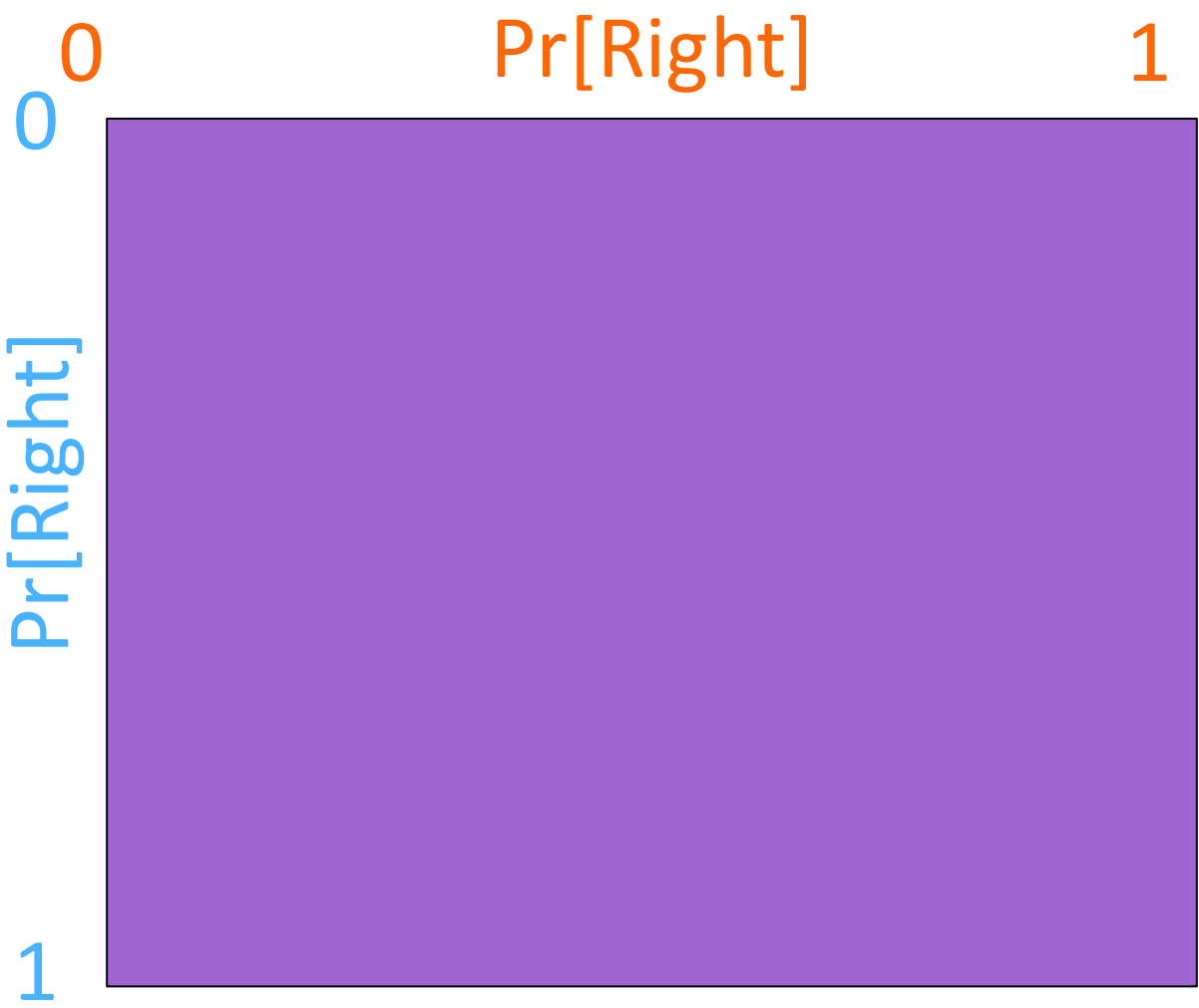
Penalty Shot Game

$f: [0,1]^2 \rightarrow [0,1]^2$, continuous
such that
fixed points $\equiv$ Nash eq.

Visualizing Nash's Proof

Kick Dive	Left	Right
Left	1 , -1	-1 , 1
Right	-1 , 1	1 , -1

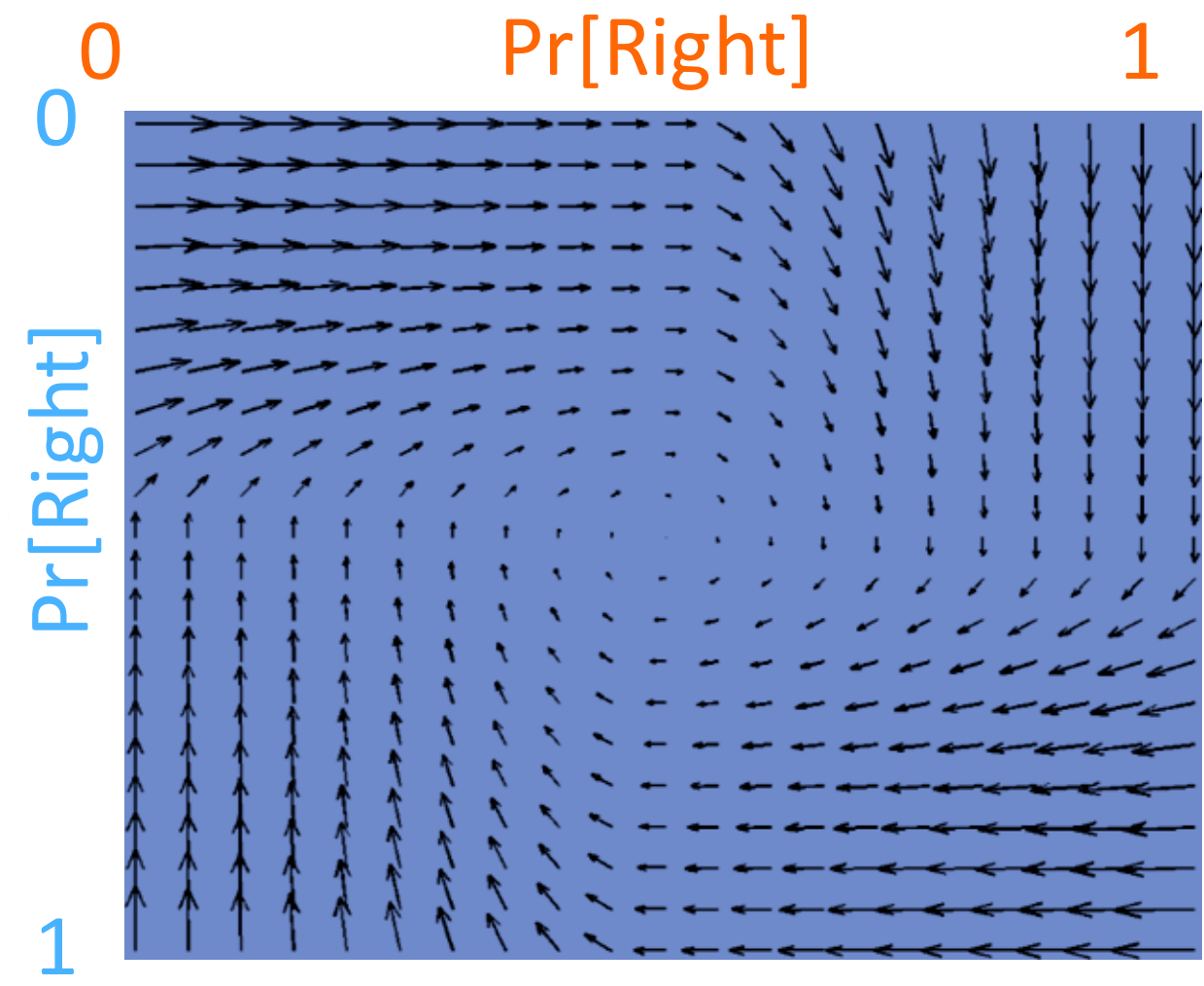
Penalty Shot Game



Visualizing Nash's Proof

Dive \ Kick	Left	Right
	Left	Right
Left	1, -1	-1, 1
Right	-1, 1	1, -1

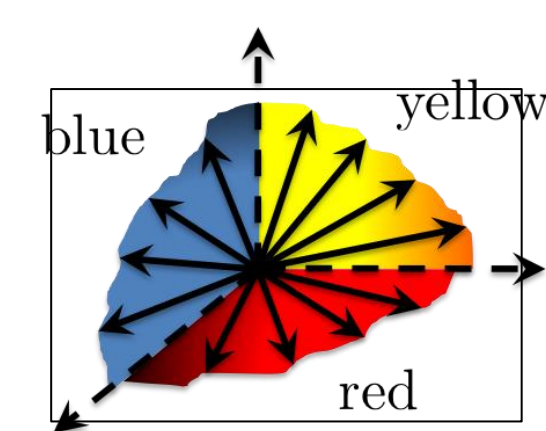
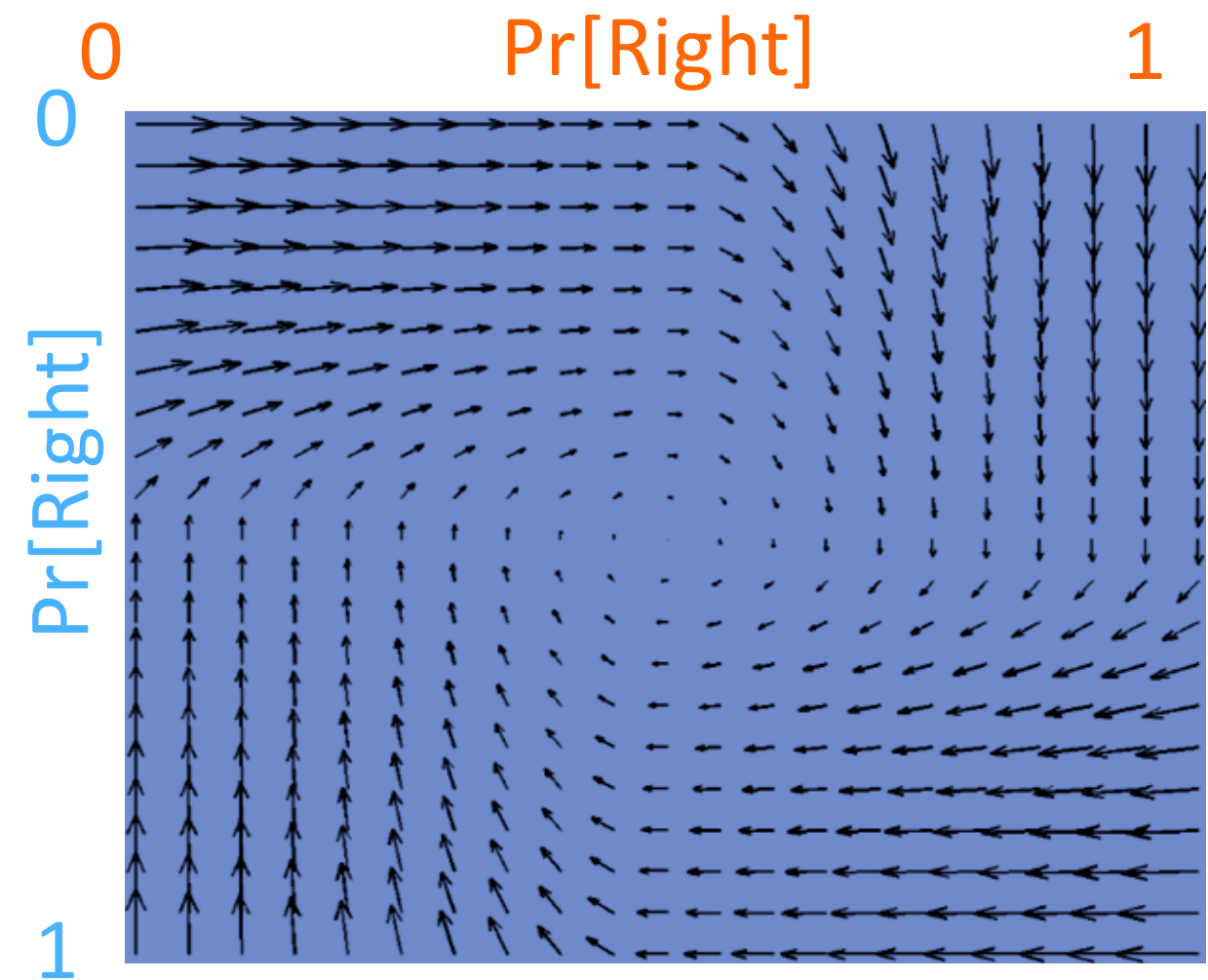
Penalty Shot Game



Visualizing Nash's Proof

Dive \ Kick	Left	Right
	Left	Right
Left	1 , -1	-1 , 1
Right	-1 , 1	1 , -1

Penalty Shot Game

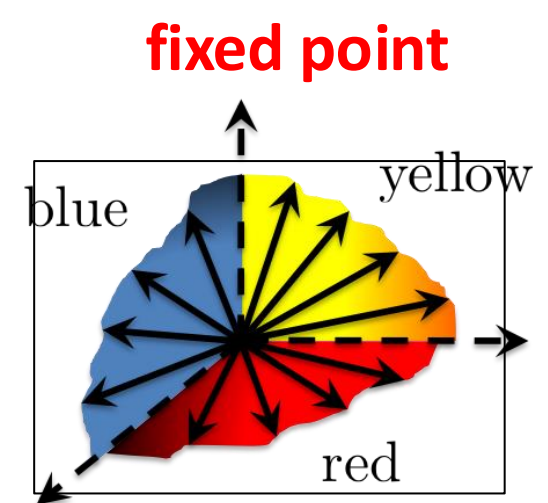
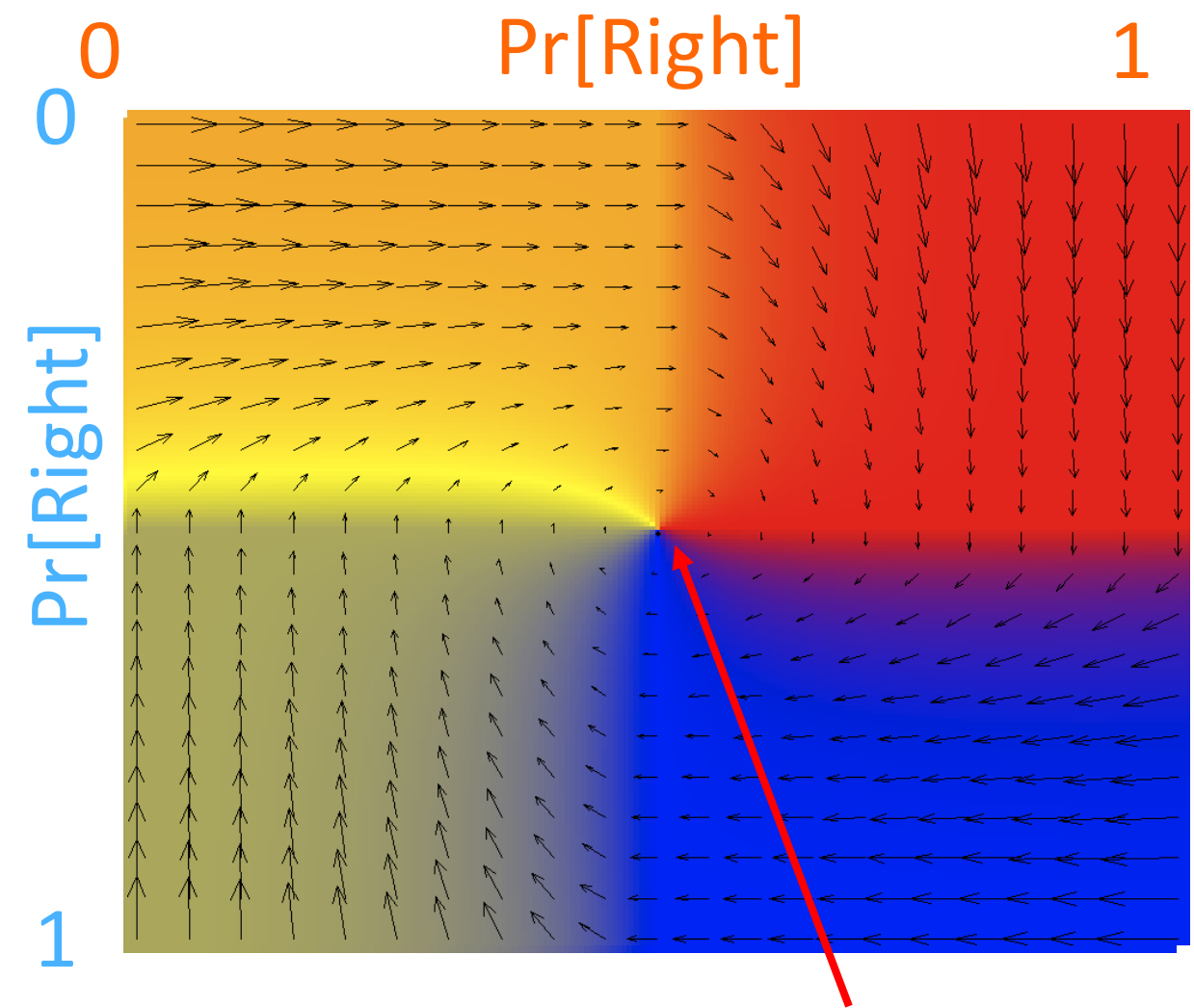


Visualizing Nash's Proof

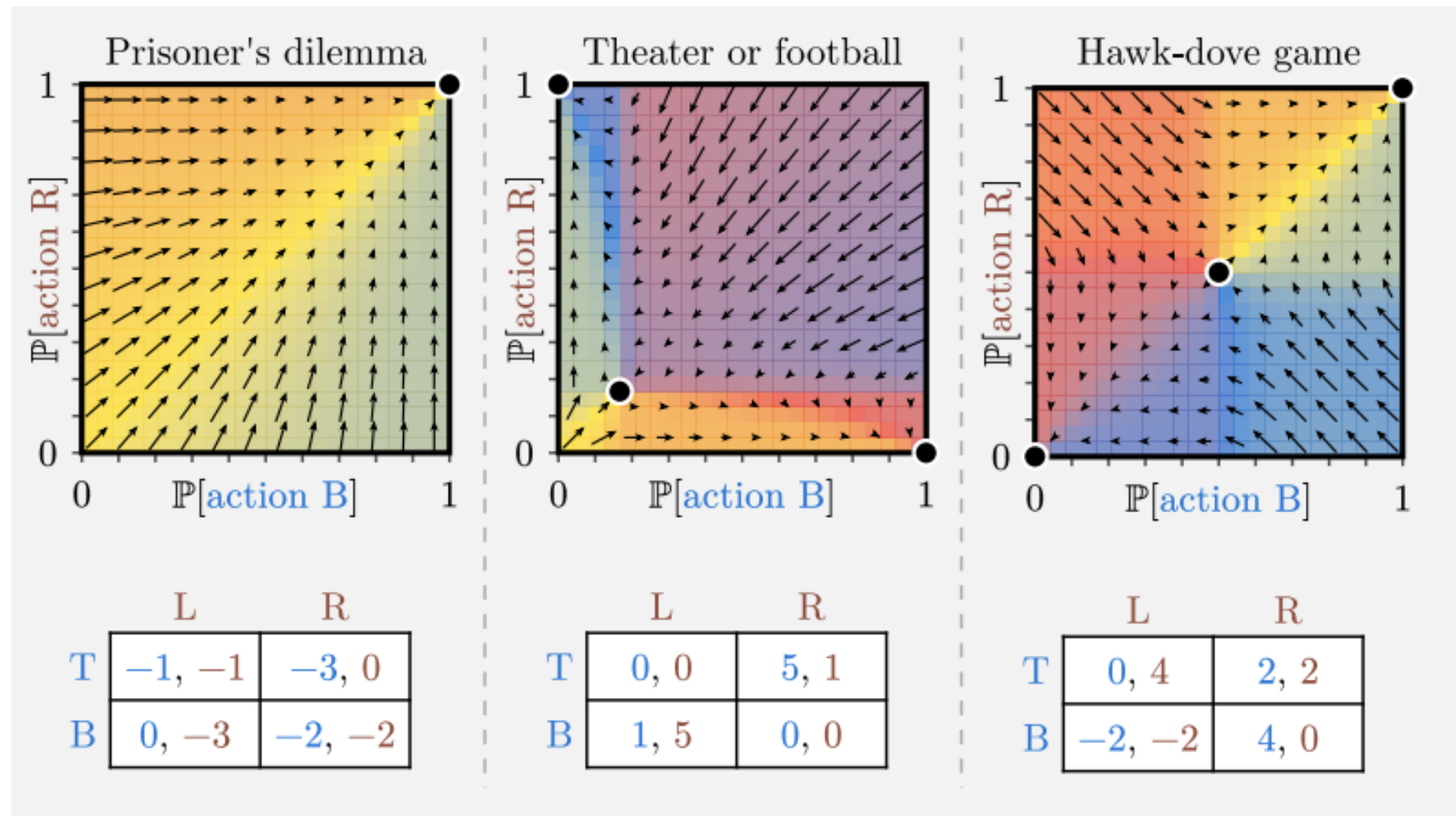
		$\frac{1}{2}$		$\frac{1}{2}$	
		Kick		Left	Right
$\frac{1}{2}$	Dive				
	Left	1, -1		-1, 1	
$\frac{1}{2}$	Right	-1, 1		1, -1	

Penalty Shot Game

Real proof: on the “board”
(and in the notes)



Other examples of Nash's Function



See notes for more examples!